

BRAINHACK 2026

RoboVerse Supplementary 2 -
Visual Inertia Odometry

AGENDA

Main goal: To provide
understanding on using VIO
with PX4

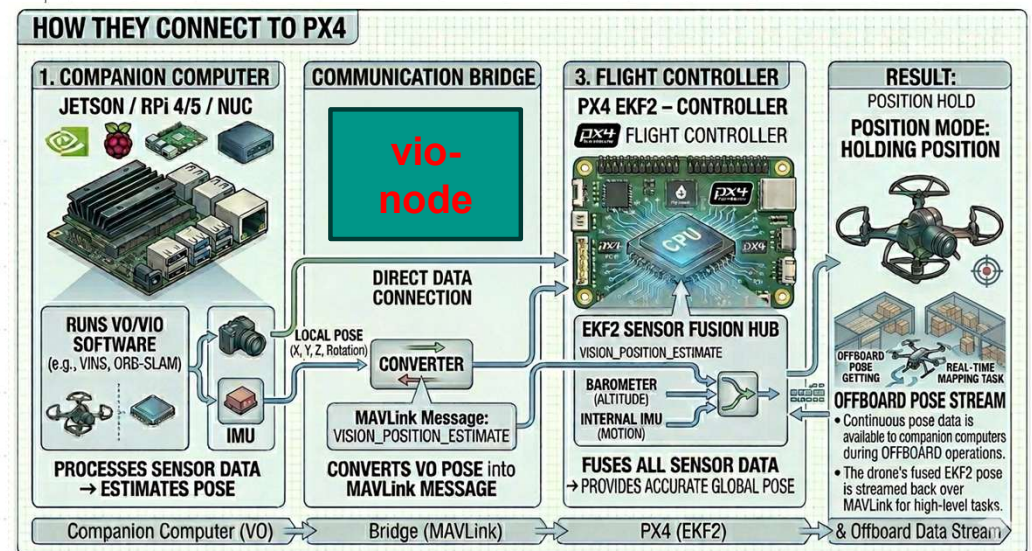
1. How does VIO work? What does VIO need?
2. How does PX4 uses VIO?

A decorative background featuring a dark grid with various symbols including plus signs, minus signs, dots, squares, and diamonds in light blue and white. The symbols are arranged in patterns, some forming larger shapes like a triangle on the left and a cluster on the right.

1. How does VIO work? What does VIO need?

How does VIO work?

- Visual inertia odometry (VIO) estimates drone position and orientation by combining camera and IMU data.
- OpenVINS is used to perform the VIO function.
- The vio_node feeds camera data and IMU data to OpenVINS.
- vio_node grabs camera data from gazebo camera (if x500 virtual drone), or Realsense (if physical drone).
- vio_bridge grabs imu from gazebo or PX4 flight controller directly.



What is meant by camera data?

- VIO is only interested on features in the image not everything in the image.
- Features are some distinct visual points in the image – sharp changes in brightness, corners and edges.
 - Plain feature environment might not be so good.
- Some popular feature extraction algorithms are SIFT, ORB, Harris, Shi-Tomasi. Opencv has libraries to do these algorithm.

```
import numpy as np
import cv2 as cv

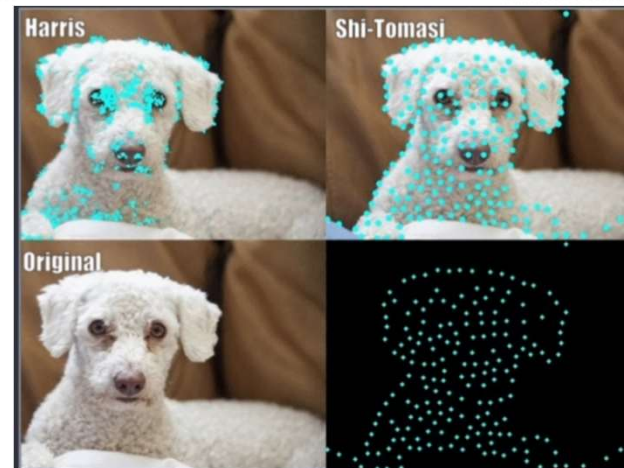
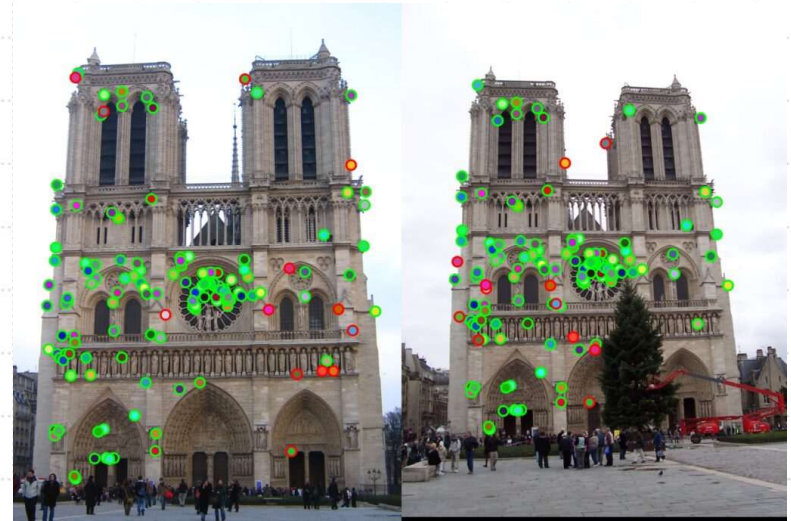
filename = 'chessboard.png'
img = cv.imread(filename)
gray = cv.cvtColor(img,cv.COLOR_BGR2GRAY)

gray = np.float32(gray)
dst = cv.cornerHarris(gray,2,3,0.04)

#result is dilated for marking the corners, not important
dst = cv.dilate(dst,None)

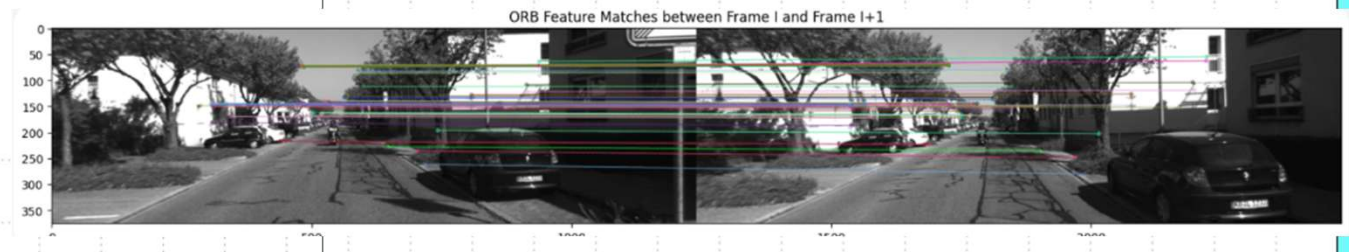
# Threshold for an optimal value, it may vary depending on the image.
img[dst>0.01*dst.max()]=[0,0,255]

cv.imshow('dst',img)
if cv.waitKey(0) & 0xff == 27:
    cv.destroyAllWindows()
```



What does VIO do with the feature?

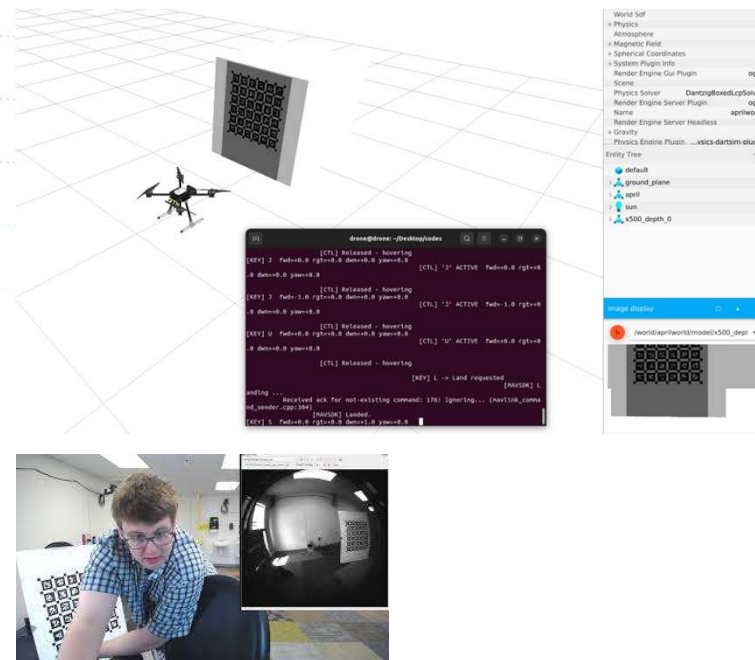
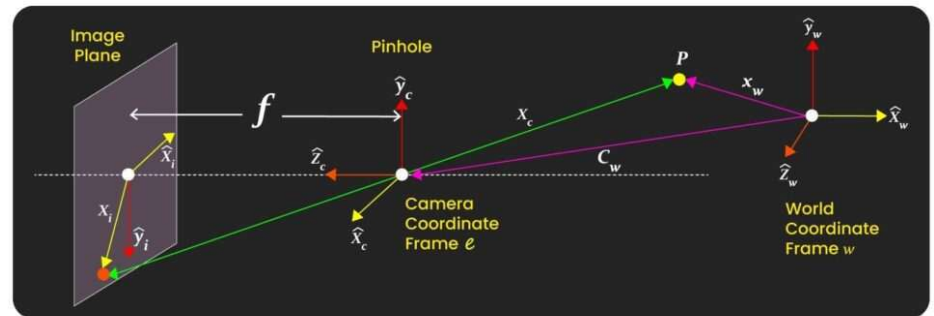
- VIO algorithm matches and track features across frames to analyse how those features move.
- Sparse Optical Flow algorithms like Lucas-Kanade method (LK)
 - A popular local method that solves the brightness constraint over a small 3x3 or 5x5 window around a pixel, assuming constant motion in that neighborhood. It is fast and suitable for real-time tracking (often used with Shi-Tomasi corner detection)
- OpenCV has libraries that does optical flow calculation.
- In other words, movement is important. **Movement of camera is especially needed during initialization to trigger calculation. Often we move the drone with the camera by hand.**



```
1 lk_params = dict(winSize=(15, 15),
2
3 maxLevel=2,
4 criteria=(cv2.TERM_CRITERIA_EPS | cv2.TERM_CRITERIA_COUNT, 10, 0.03))
5
6 # Calculate optical flow
7 points2, status, err = cv2.calcOpticalFlowPyrLK(image1, image2, points1, None, **lk_params)
8
9 # Select good points
10 good_new = points2[status.ravel() == 1]
11 good_old = points1[status.ravel() == 1]
```

IMU data and Pose Estimation

- A 3D object is represented onto the Image Plane (2D image) using Extrinsic Matrix and Intrinsic Matrix:
 - Extrinsic Matrix: It is external to the camera describing transformation with the World and camera.
 - Intrinsic Matrix: It is internal to the camera describing transformation from camera coordinate to image plane.
- IMU data is fused with position estimation to determine the position of the camera based on the movement.
 - IMU data needs to transform to camera frame and from camera frame back using extrinsic matrix, to fuse with position and movement estimation.
- Intrinsic Matrix is used for epipolar geometry calculation.
- OpenVINS and other VIO algorithms need to be provided with values for extrinsic matrix (transformation between IMU and camera) and intrinsic matrix.
- IMU-Camera calibration needs to be performed to determine these matrix value.
 - Calibration uses a 8 x 8 April tag board as the known distance for the software to determine the values.
 - Refer to <https://docs.openvins.com/gs-calibration.html> if you like to know how it is being done.
 - **You do not need to perform this calibration process for the Final Challenge drone. But need to be aware that placement of camera on the drone must not be changed as it will invalidate the calibration.**



1. How does PX4 uses VIO?

How does PX4 uses VIO?

- PX4 can use VIO instead of GPS as a source of
 - Horizontal position
 - Vertical position
 - Velocity
- You need to configure it.
- PX4 uses EKF2 to fuse the inputs with IMU data and barometer data, to provide the final position estimation.
- During startup, PX4 will wait for position estimate from VIO (vio-node). If not received, EKF2 will not be ready and PX4 unable to arm.
- As such, **it is important to make sure vio-node is running and physically take up the drone to do slight movement along all axis of the drone to trigger VIO initialization. Perform until the initialization is successful.**
- When initialization is successful, vio-node will output position estimate.

How to assess VIO position is ok?

- Observe the vio-node position estimate:
 - If the position deviates too much from ground truth when you move, bad calibration.
 - Solution: Restart and redo.
 - If the position estimate drift infinitely, it is no good due to bad physical initialization. Wrong extrinsic and intrinsic.
 - Solution: Recalibrate.
 - If stuck at waiting for imu data, imu source not ready.
 - Solution: Check if wire between flight controller and offboard is intact.

References

- Those who wish to start on programming your VIO, this link is a good starting point:
 - <https://learnopencv.com/monocular-slam-in-python/>
- Know more about OpenVins:
 - <https://www.youtube.com/watch?v=dXN2E38jvQM&t=294s>

THANK YOU